

Scan QR Code For Attendance



Micromouse

Lecture 1: Introduction to MM

October 16

Meet the Board!



Nick Ward



Eaton Huang



Lucas Zhong



Colton Kaneria



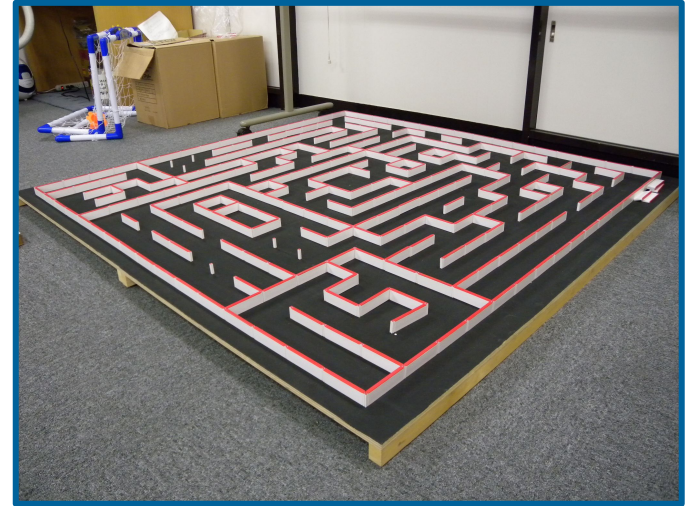
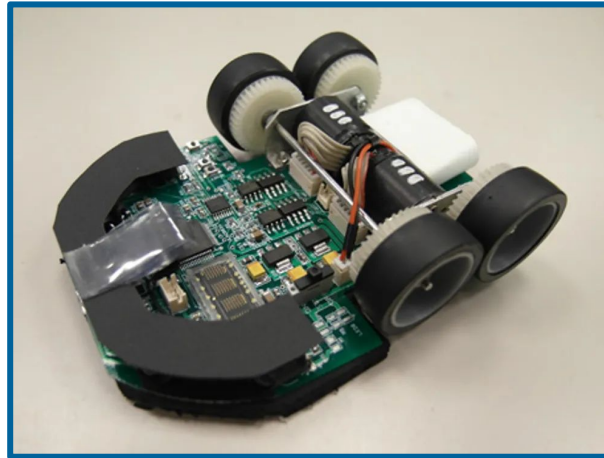
Adam Wu

Justin Chen

Jeremiah Yong

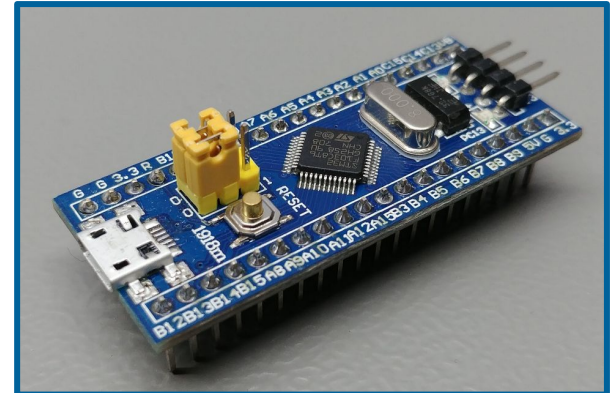
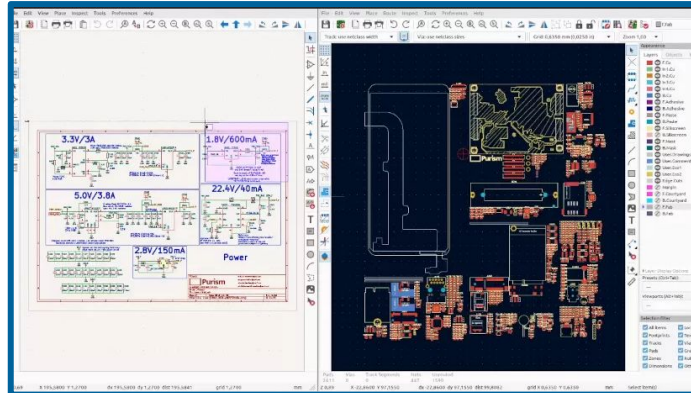
What is Micromouse?

- International robotics competition
- Develop an autonomous robotic “mouse” to navigate and solve mazes



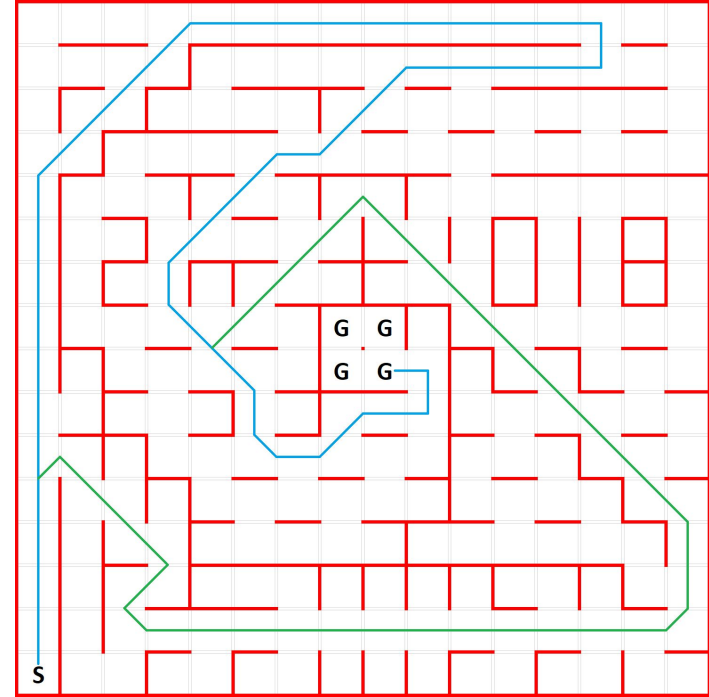
What will you learn?

- Embedded Systems
- C/C++
- Soldering
- Circuit Analysis
- Hardware Debugging
- Microcontrollers
- Git



Goals

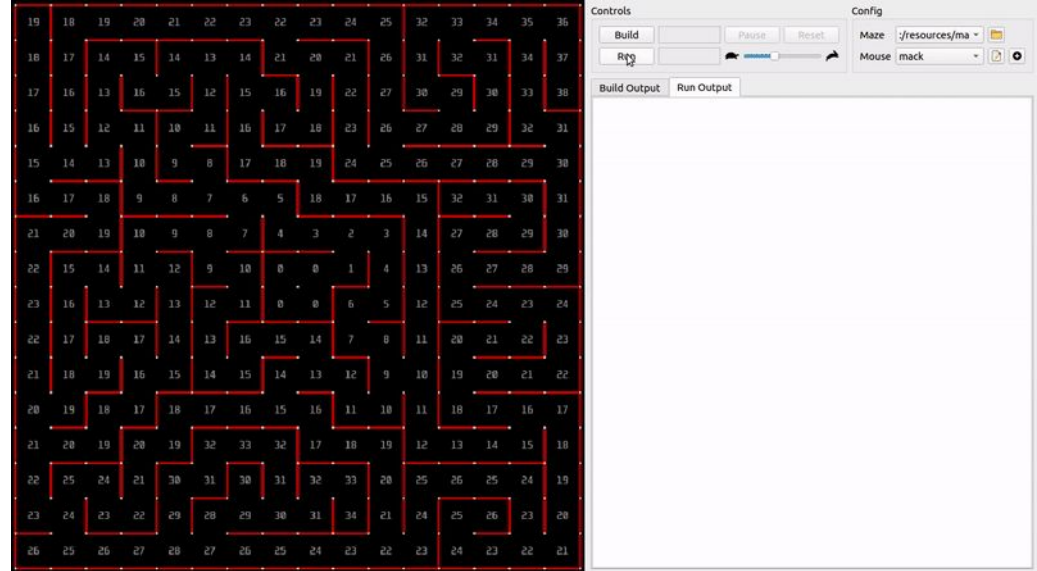
- To have all teams compete in AAMC in Spring 2025
- Understand basic embedded system hardware design
- Understand basic PCB design
- Understand basic embedded programming
- *Have fun!* (it's a lot of fun)



Software Overview

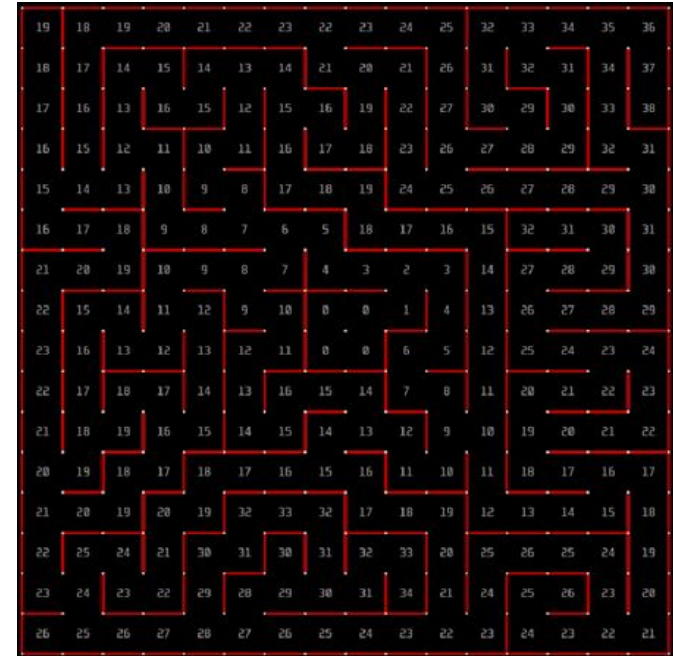
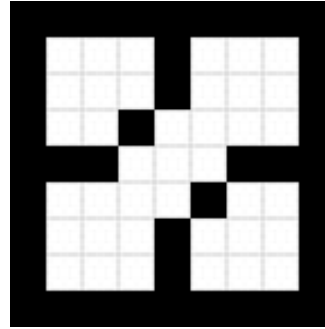
Simulator

- Software portion of project
- Open Source software
- Implement data structures and necessary functions
- Learn flood-fill and basic mouse movements



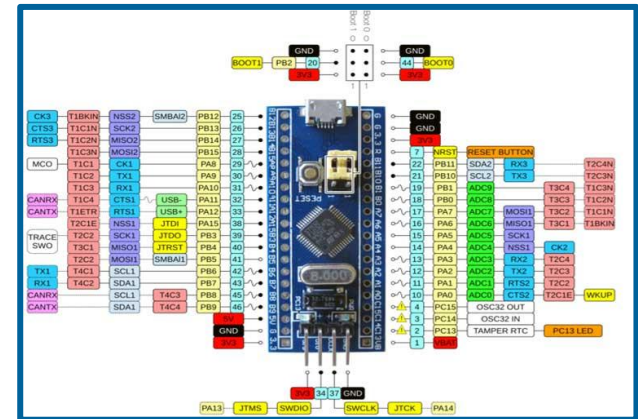
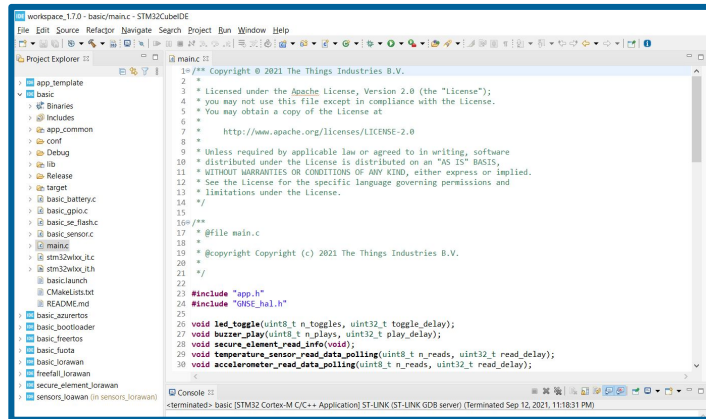
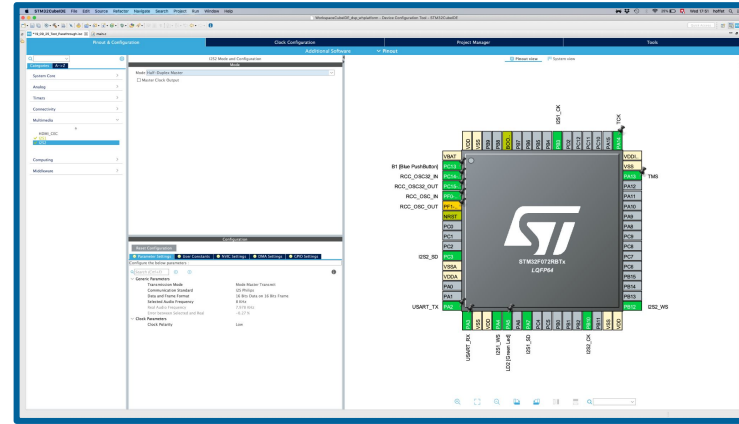
Search/Navigation Algorithm

- Implementation of Floodfill
 - Searching as much of maze as possible
 - Updates distances + walls while traversing through maze
- Finding shortest path(s)



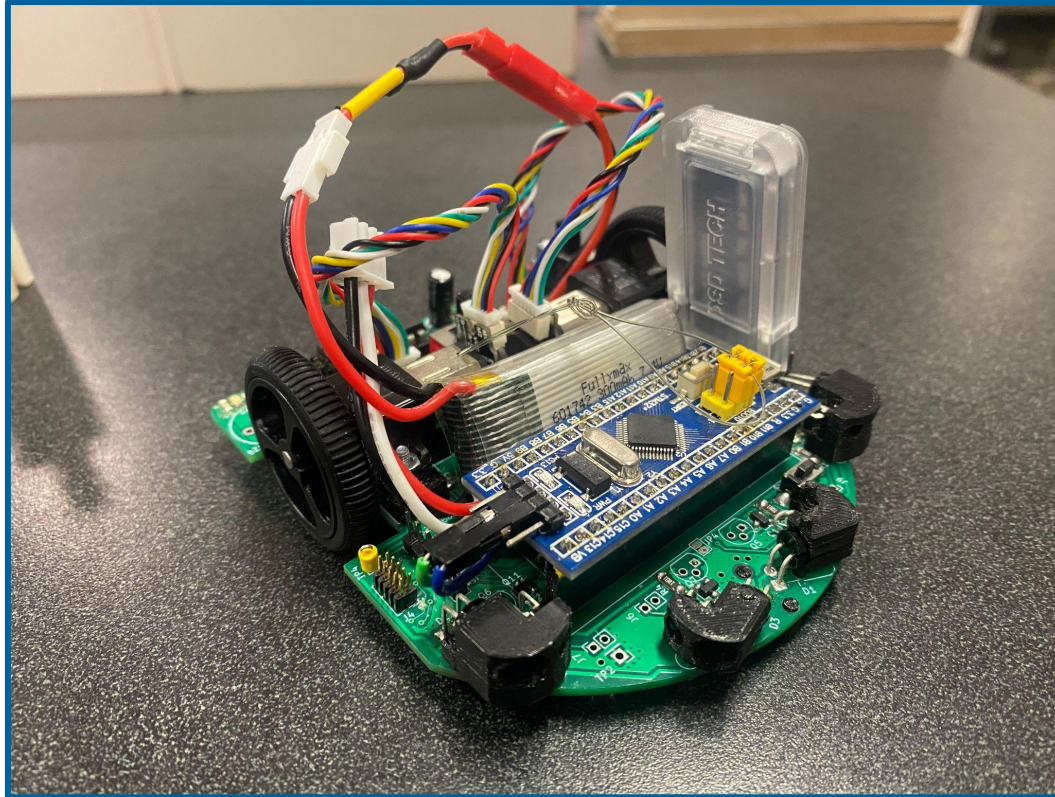
STM32 CubeIDE

- Development environment IDE by STMicroelectronics
- Implement pin configuration and management
- Peripheral configuration



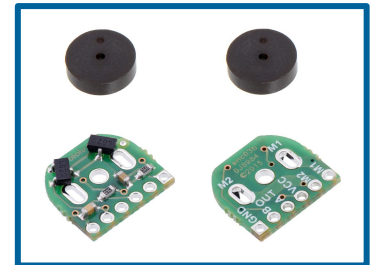
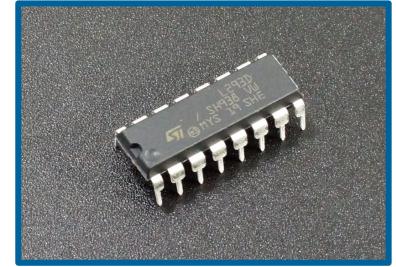
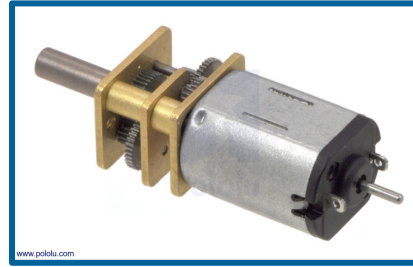
Hardware Overview

Design Example



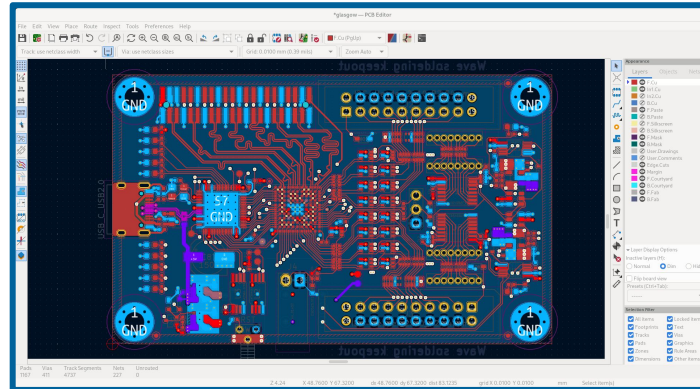
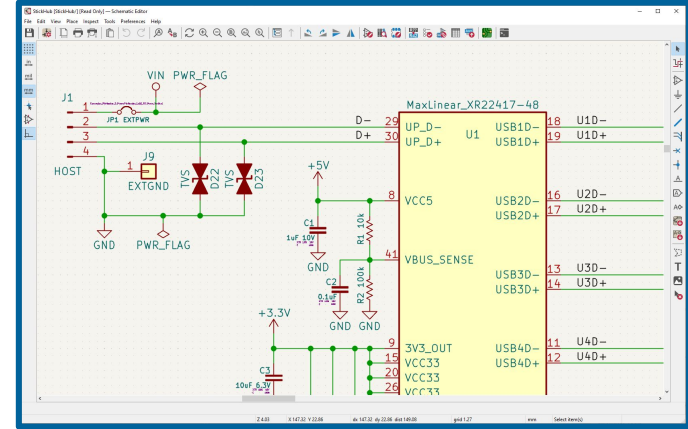
Components

- Motors
- Motor Driver
- Microcontroller
- IR Sensors
- Battery
- Encoders



Kicad 8

- Schematic building
- PCB design
- Free software
- <https://www.kicad.org/download/>



UROP (Undergraduate Research Opportunities Program)

- Fall Deadline: early November
- Free funding for Micromouse
- Requirements:
 - An advisor
 - Research Proposal
 - Personal Statement
 - Faculty Mentor Agreement Form

<https://uop.uci.edu/>

*more information on last slide!



Course Outline

Fall: Hardware Design & Algorithm

- Understanding MM schematic
- Designing MM PCBA
 - Routing in KiCAD
- Floodfill
 - MM simulator
 - navigation algorithms
- Team Formation

Winter: Hardware Assembly & Firmware

- Soldering designed PCBA
- Firmware to interface with motors, sensors, etc.
 - Ex. PID
- Integrating firmware with floodfill

Spring: Preparation for Competition

- Software optimization
- Catch up workshops
- Finalizing the mouse
- AAMC

Fall Schedule

Week	Meeting Date	Meeting Time	Meeting Location
3	10/16	1:00-3:00 pm	EH2430
4	10/23	1:00-3:00 pm	EH 2430
5	10/30	1:00-3:00 pm	EH 2430
6	11/6	1:00-3:00 pm	EH 2430
7	11/13	1:00-3:00 pm	EH 2430
8	11/20	1:00-3:00 pm	EH 2430
10	12/4	1:00-3:00 pm	EH 2430

Additional Resources

Extra Micromouse Links

- [Brief History of Micromouse](#)
- [Micromouse Rules](#)
- [Micromouse Website](#)

UROP Details

- [UROP Website](#)
- [UROP Info](#)
- [Sample UROP Proposals](#)

IEEE Information

- [IEEE Discord](#)
- [IEEE Website](#)



Template Slide

